

Econ 101A

Section 18

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1 Bertrand Duopoly

- In Bertrand's game, each firm chooses a price, and produces enough output to meet the demand it faces, given the prices chosen by all the firms.
- Assumptions: A single good is produced by two firms ($i = 1, 2$); each firm can produce q_i units of the good at a cost of $C_i(q_i) = C(q_i)$.
- The interpretation of demand function D in this economic setting is that if the good is available at the price p , then the total amount demanded is $D(p)$.
- Assume that if the firms set different prices, then all consumers purchase the good from the firm with the lowest price, which produces enough output to meet this demand. If more than one firm sets the lowest price, all the firms doing so share the demand at that price equally. A firm whose price is not the lowest price receives no demand and produces no output.
- Thus, the profit function of firm i can be written as

$$\pi_i(p_i, p_j) = \begin{cases} p_i D(p_i) - C(D(p_i)) & \text{if } p_i < p_j \\ p_i \frac{D(p_i)}{2} - C\left(\frac{D(p_i)}{2}\right) & \text{if } p_i = p_j \\ 0 & \text{if } p_i > p_j \end{cases}$$

- We can find the Nash equilibria of the game by finding the firms' best response functions.
- Let's assume $C_i(q_i) = cq_i$ for $i = 1, 2$ and assume that the demand function is linear and it's given by $D(p) = \alpha - p$ for $p \leq \alpha$ and $D(p) = 0$ for $p > \alpha$, where $\alpha > 0$ and $c \geq 0$ are constants.
- Then firm i 's best response function is given by

$$BR_i(p_j) = \begin{cases} \{p_i : p_i > p_j\} & \text{if } p_j < c \\ \{p_i : p_i \geq p_j\} & \text{if } p_j = c \\ \emptyset & \text{if } c < p_j \leq p^m \\ \{p^m\} & \text{if } p^m < p_j \end{cases}$$

- Explanation:
 - If $p_j < c$, then firm i 's profit is negative if $p_i \leq p_j$ and zero if $p_i > p_j$. Thus any price greater than p_j is a best response to p_j .

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- If $p_j = c$, then any price greater than or equal to p_j yields a profit of zero, and hence is a best response to p_j .
- If $c < p_j \leq p^m$, then there is no best response. Firm i wants to choose a price less than p_j , but is better off the closer that price is to p_j . For any price less than p_j there is a higher price that is also less than p_j , so there is no best price.
- If $p^m < p_j$, then p^m (the monopoly price) is the unique best response of firm i .
- A Nash equilibrium is a pair (p_1^*, p_2^*) of prices such that p_1^* is a best response to p_2^* , and p_2^* is a best response to p_1^* .
- The above described Bertrand duopoly has a unique Nash equilibrium, in which each firm's price is equal to marginal cost c , that is, $(p_1^*, p_2^*) = (c, c)$.

2 Cournot Duopoly

- In Cournot's game, each firm chooses an output, the price is determined by the demand for the good in relation to the total output produced.
- Assumptions: A single good is produced by two firms ($i = 1, 2$); each firm can produce q_i units of the good at a cost of $C_i(q_i) = C(q_i)$.
- All the output is sold at a single price, determined by the demand for the good and the firms' total output. Specifically, if the firms' total output is Q then the market price is $P(Q)$; P is called the "inverse demand function."
- Assume that P is a decreasing function when it is positive: if the firms' total output increases, then the price decreases (unless it is already zero). If the output of each firm i is q_i , then the price is $P(q_i + q_j)$, so that firm i 's revenue is $q_i \cdot P(q_i + q_j)$.
- Thus, the profit function of firm i can be written as $\pi_i(q_i, q_j) = q_i \cdot P(q_i + q_j) - C_i(q_i)$.
- We can find the Nash equilibria of the game by finding the firms' best response functions.
- Let's assume $C_i(q_i) = cq_i$ for $i = 1, 2$ and assume that the inverse demand function is linear and it's given by $P(Q) = \alpha - Q$ for $Q \leq \alpha$ and $P(Q) = 0$ for $Q > \alpha$, where $\alpha > 0$ and $c \geq 0$ are constants.
- Then firm i 's best response function is given by

$$BR_i(q_j) = \begin{cases} \frac{1}{2}(\alpha - c - q_j) & \text{if } q_j \leq \alpha - c \\ 0 & \text{if } q_j > \alpha - c \end{cases}$$

- Explanation:
 - If $q_j \leq \alpha - c$, then the output that maximizes firm i 's profit is $q_i = \frac{1}{2}(\alpha - c - q_j)$
 - If $q_j > \alpha - c$, the value of $(\alpha - c - q_j)$ is negative, thus for such a value of q_j , firm i 's revenue will be negative for all positive values of q_i . This means that firm i 's profit is negative for any positive output, so its best response is to produce the output of zero.
- A Nash equilibrium is a pair (q_1^*, q_2^*) of outputs such that q_1^* is a best response to q_2^* , and q_2^* is a best response to q_1^* .
- The above described Cournot duopoly has a unique Nash equilibrium, in which each firm's output is equal to $\frac{1}{3}(\alpha - c)$, that is, $(q_1^*, q_2^*) = (\frac{1}{3}(\alpha - c), \frac{1}{3}(\alpha - c))$.

3 Exercise

Initially there is one firm in a market for cars. The firm has a linear cost function: $C(q) = 2q$. The market inverse demand function is given by $P(Q) = 9 - Q$.

1. What price will the firm charge? What quantity of cars will the firm sell? How much profit will the firm make?
2. Now, a second firm enters the market. The second firm has an identical cost function. Suppose the firms compete via Bertrand competition. What will be the equilibrium output of each firm? What will be the equilibrium profit of each firm?
3. Suppose instead the firms compete via Cournot competition. What will be the equilibrium output of each firm? What will be the equilibrium profit of each firm?
4. Which type of market do consumers prefer: Monopoly, Bertrand duopoly, or Cournot duopoly? Why?